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Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology
School of Computing

Department of Computer Science and Engineering

10211CS224 / FULL STACK APPLICATION DEVELOPMENT (JAVA)

PROJECT REVIEW : MILESTONE PROJECT – 1

Date: 25-03-2026

Dynamic Online Examination & Result Processing System

SDG : SDG - 4 Quality Education

Presented By:

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Courses Handling Faculty:

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INTRODUCTION



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- Traditional examination systems often rely on manual paper-based processes.
- Manual evaluation consumes time and increases the risk of human error.
- Managing student records and results manually reduces efficiency.
- There is a growing need for digital transformation in academic assessments.
- This project introduces a fully automated online examination system.
- It enables secure login, dynamic question display, and automatic evaluation.
- Results are generated instantly and stored in a database.
- The system improves speed, accuracy, and transparency in examinations.

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PROBLEM STATEMENT



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- Many educational institutions still conduct exams manually or with partially automated systems.
- Manual result processing leads to delays and calculation errors.
- Managing large volumes of student data becomes difficult and inefficient.
- There is no centralized platform for secure exam management.
- Existing systems lack automation in evaluation and reporting.
- A reliable, secure, and scalable online examination system is required.
- The system must reduce human intervention and improve accuracy.
- It should provide fast result processing and structured data management.

OBJECTIVE(S)



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- To design and develop a web-based Dynamic Online Examination and Result Processing System with secure authentication and automated evaluation.
- To implement database-driven exam management that improves efficiency, accuracy, and performance of result processing.

PROPOSED SYSTEM



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- Provides secure login for students and administrators.
- Allows administrators to create exams and manage questions dynamically.
- Displays randomized questions during examination.
- Implements timer-based exam submission.
- Automatically evaluates answers and calculates marks.
- Stores results securely in a MySQL database.
- Provides quick access to previous results and performance records.
- Reduces manual workload and increases reliability.

PROPOSED SYSTEM ARCHITECTURE



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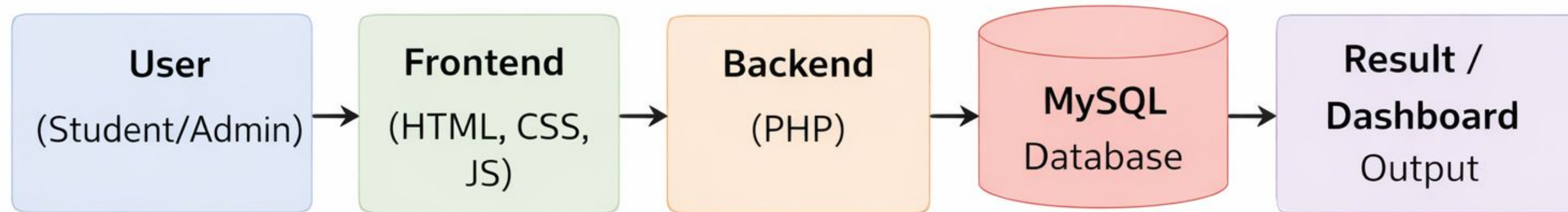
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WORKFLOW



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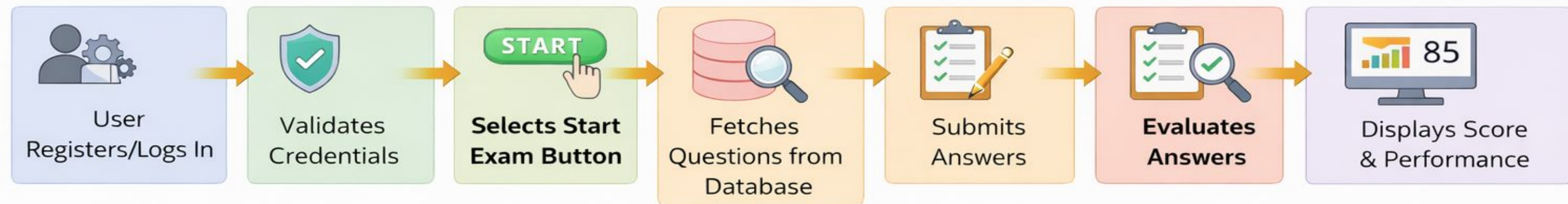
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TECHNOLOGIES USED



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Frontend:

HTML5, CSS3

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Backend:

PHP

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Database:

MySQL

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Tools:

XAMPP, phpMyAdmin, VS Code

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Hardware/Software Requirements:

- Minimum 4GB RAM
- Windows/Linux OS
- Web Browser (Chrome/Edge)
- Apache Server (via XAMPP)

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MODULE DESCRIPTION



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Module 1: User Management

Handles registration, login, authentication, and session control securely.

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Module 2: Examination Management

Allows dynamic question and answer retrieval from the database.

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Module 3: Result Processing

Automatically evaluates answers, calculates marks, and stores results.

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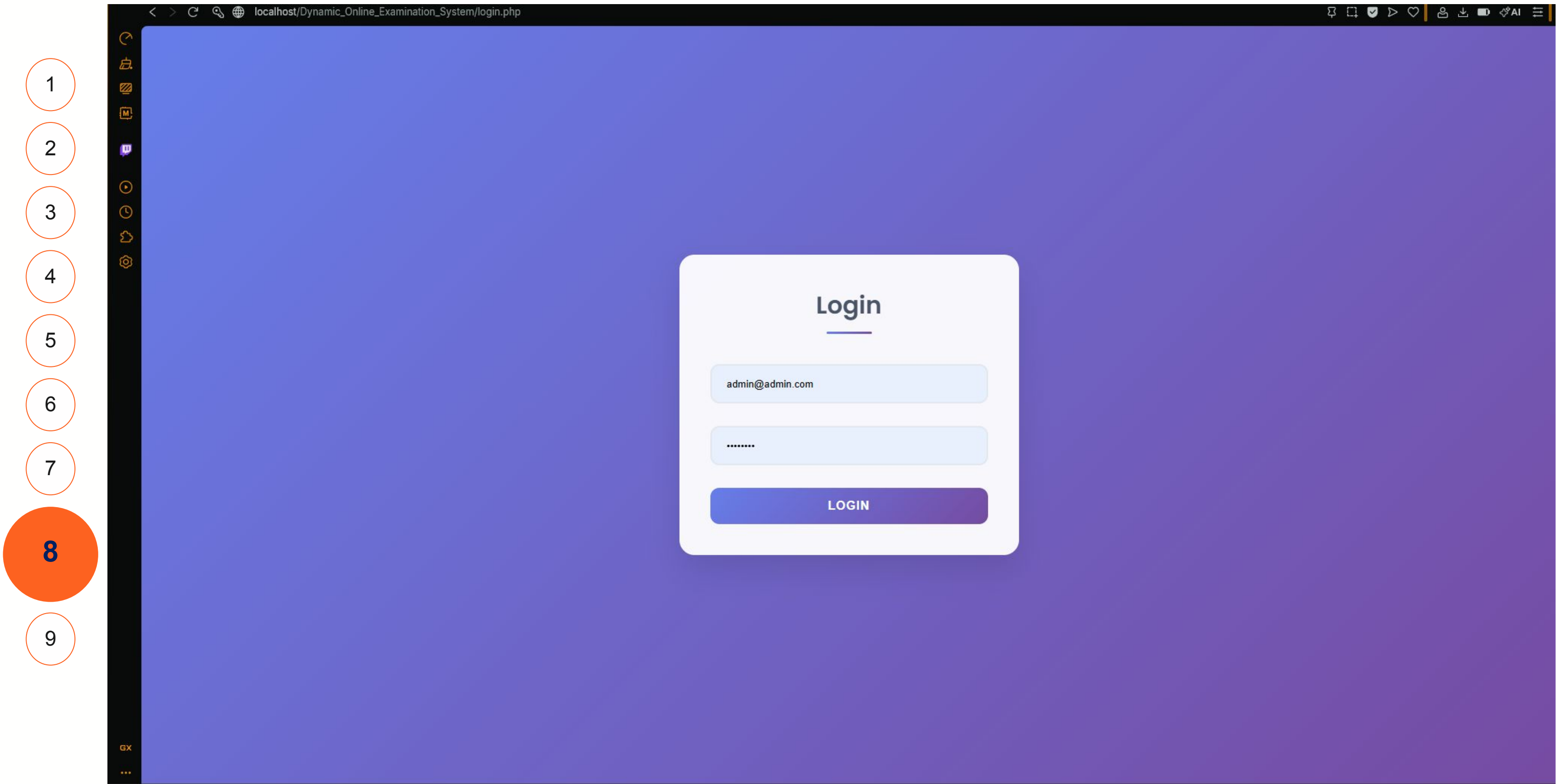
Module 4: Database Management

Manages structured storage of users, questions, and results ensuring data integrity.

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IMPLEMENTATION SCREENSHOTS



IMPLEMENTATION SCREENSHOTS



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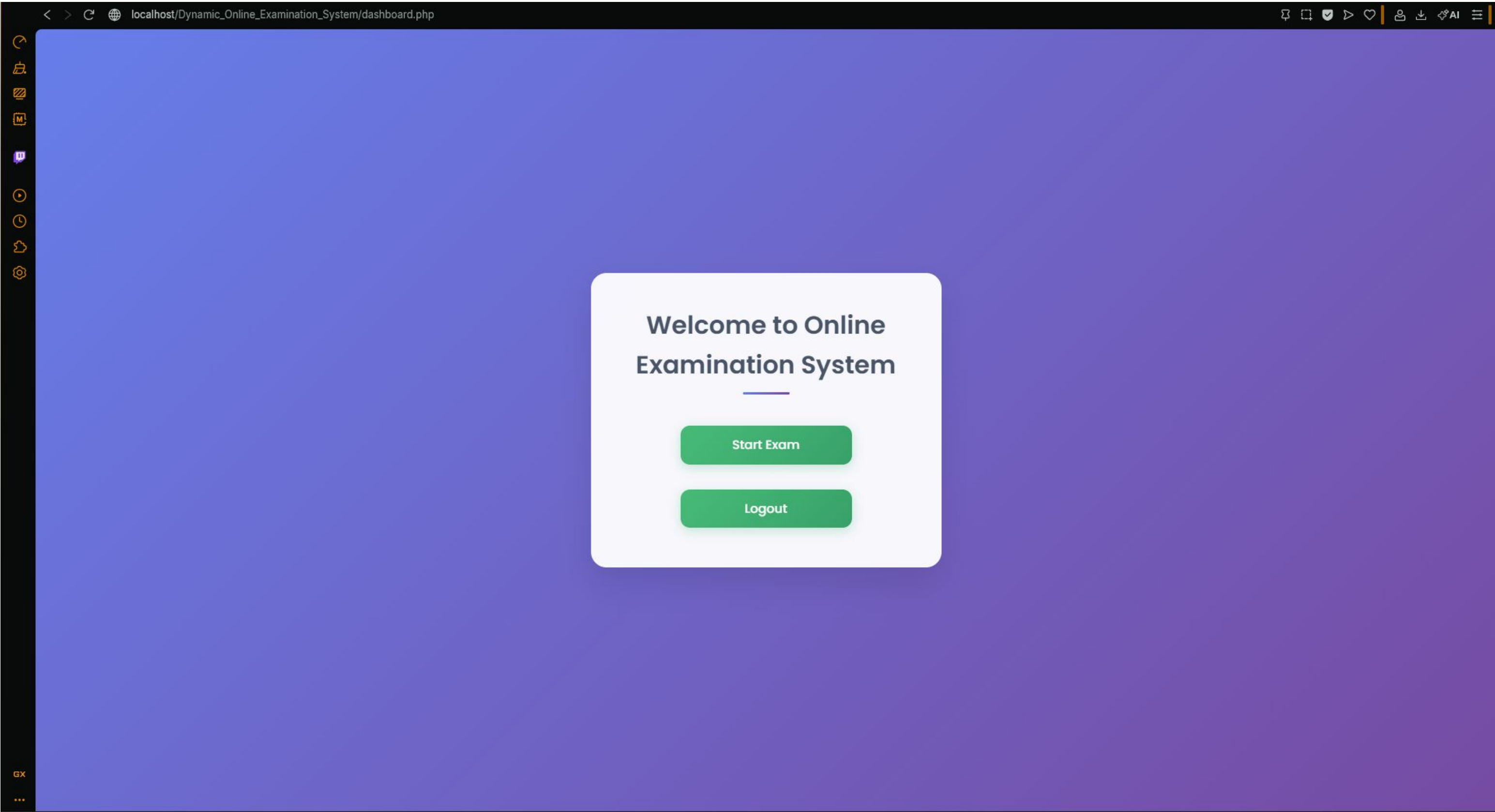
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IMPLEMENTATION SCREENSHOTS



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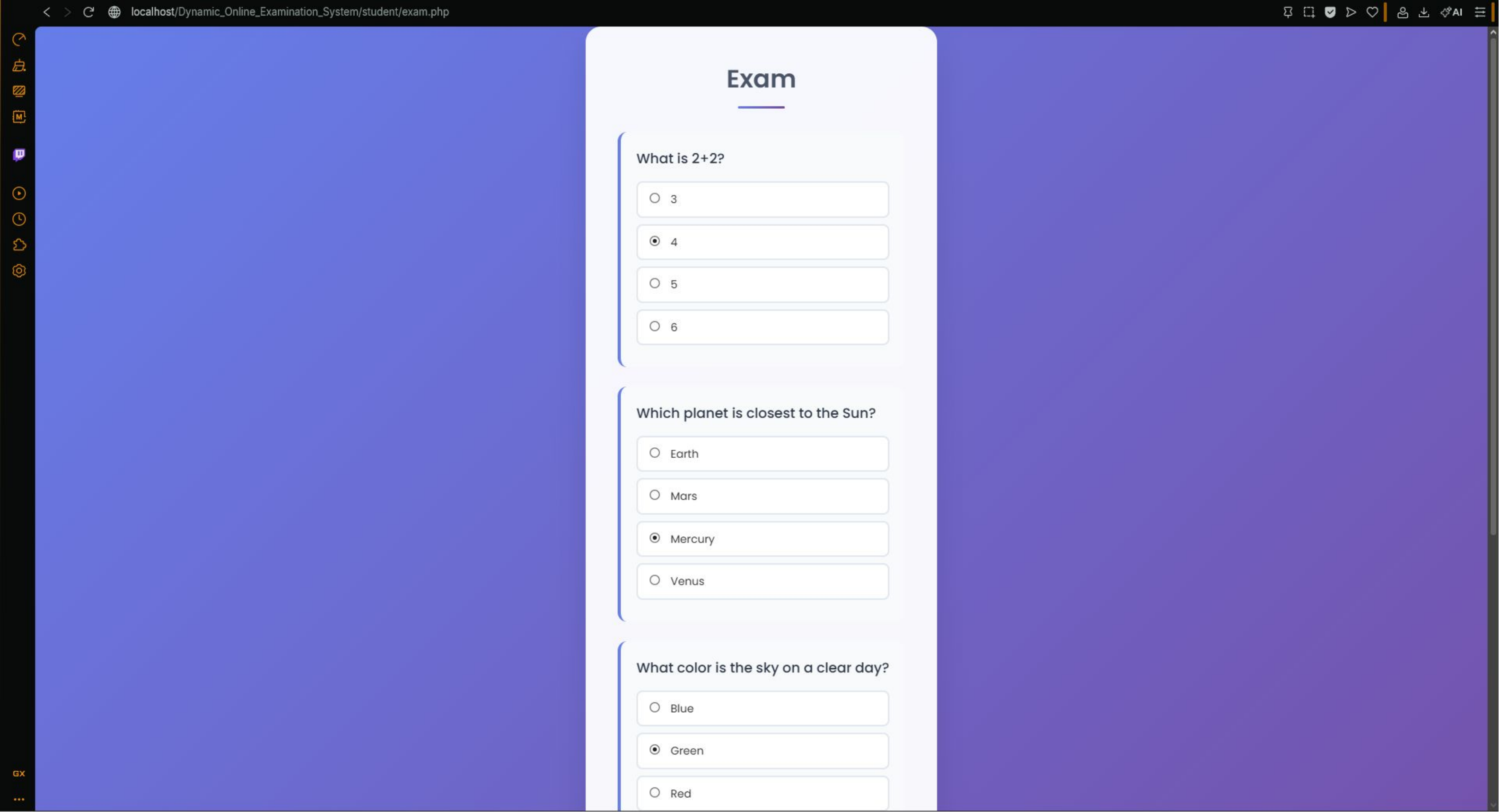
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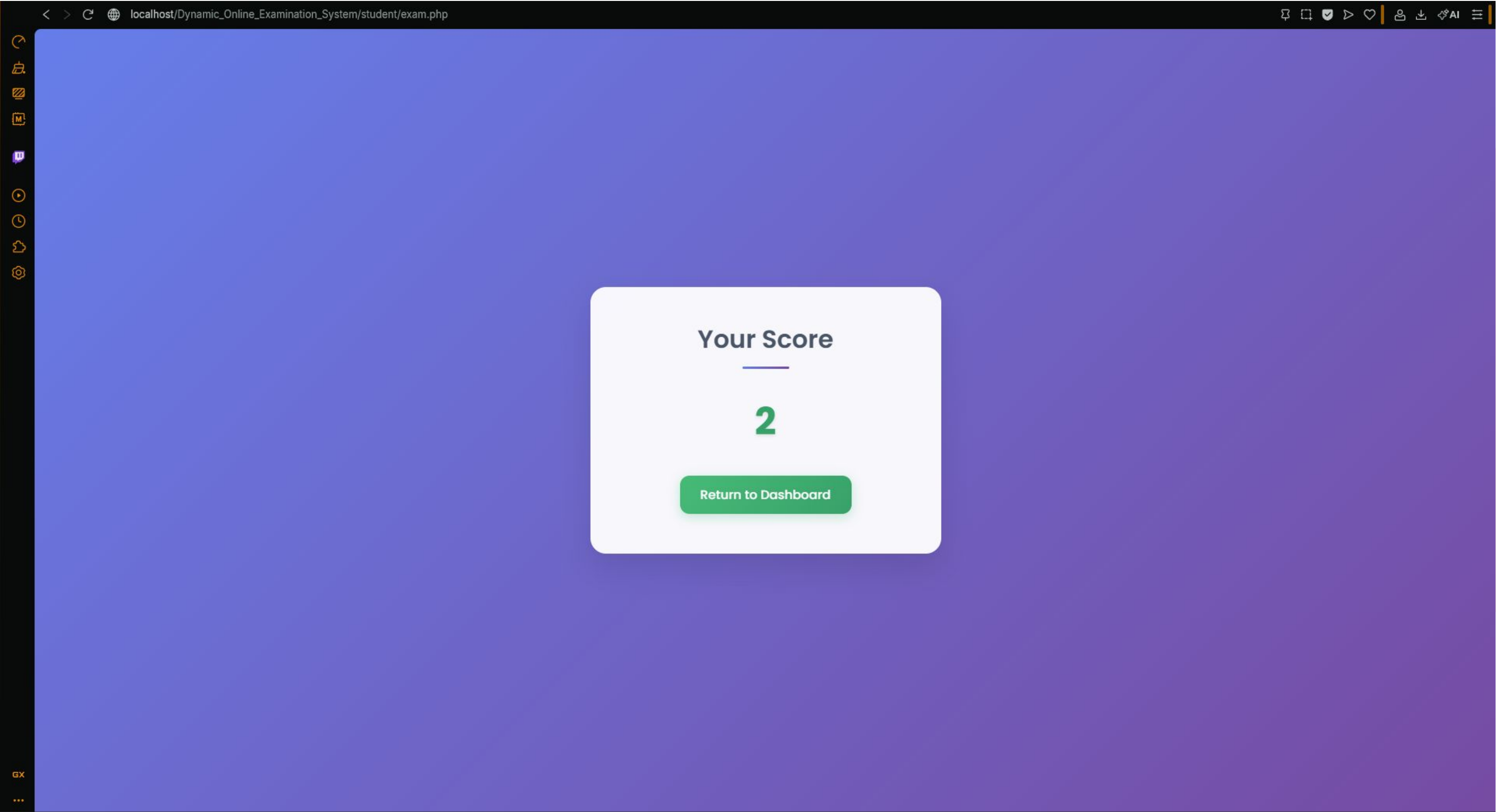
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IMPLEMENTATION SCREENSHOTS



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ADVANTAGES & LIMITATIONS



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Advantages

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- Reduces manual errors in evaluation.
- Saves time in result processing.
- Secure data storage and access control.
- Scalable for large number of students.
- Instant result generation.

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Limitations

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- Requires internet or local network access.
- Dependent on server availability.
- Basic implementation may lack advanced analytics.
- Security needs enhancement for large-scale deployment.

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Thank you

